

DEPARTMENT OF INDUSTRIAL ENGINEERING



Term Project Report

Simulation and Optimization of Cylinder Body at Ravi Autos Shiekhupura, Lahore

INDUSTRIAL SIMULATION (IE-316)

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Assessment Rubrics

Knowledge Component	Domain	PLO mapped	Taxonomy Level	Contribution	Max Marks	Obtained Marks
APPLY the simulation concepts to model systems of practical nature	Cognitive	2	C-3	67%	5	
ANALYZE the input data by using ARENA Input Analyzer to find out the best distribution fit.	Cognitive	2	C-4		5	
USE ARENA or SIMIO software to model the problem	Cognitive	5	C-3		5	
EVALUATE the performance of the simulation model to optimize the solution	Cognitive	2	C-5		5	
RESPONDING positively towards deadline for project submissions.	Affective	11	A2	33%	5	
RESPONDING the aspects of innovation in projects	Affective	12	A2		5	

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1. Introduction:

1.1. Overview of Ravi Autos:

Ravi Autos Shiekhupura Pvt. Limited is an ISO 2015 certified company established in 1981, has evolved from a humble foundry to a diversified manufacturing powerhouse. Today, they supply a wide range of tractor and motorbike Parts.

1.2. Importance of Cylinder Body:

The cylinder body is a crucial part of the motorbike engine, impacting performance, fuel efficiency, emissions, and overall motorbike reliability.

1.3. Objective of the Project:

This project aims to simulate and optimize the utilization of resources, Queue length and waiting time etc.

1.4. Overall insight:

- This report presents the findings and results of a comprehensive simulation and optimization study on the cylinder body at Ravi Autos Sheikhpura. The cylinder body is a critical component of the engine, and its design and performance have a direct impact on the overall efficiency and reliability of the vehicle. In this study, we employed advanced simulation tools and optimization techniques to investigate the current design of the cylinder body and identify opportunities for improvement. The goal of this research was to optimize the design of the cylinder body to achieve improved thermal and structural performance, reduced weight, and enhanced overall performance. The results of this study have important implications for the design and manufacture of cylinder bodies at Ravi Autos Sheikhpura, and have the potential to contribute to the development of more efficient and reliable engines.
- The simulation was conducted through **Arena Software** and optimization was conducted through optimization analysis was mainly done through **Input Analyzer**.

2. Literature Review:

The cylinder body is a critical component of an engine, and its design and performance have a direct impact on the overall efficiency and reliability of the vehicle. Simulation and optimization techniques have become increasingly important in the design and development of engine

components, allowing for the evaluation and improvement of performance without the need for physical prototypes. This literature review aims to provide a comprehensive overview of the current state of research in the simulation and optimization of cylinder bodies at Ravi Autos Sheikhpura, with a focus on specific operations such as facing, turning, boring, offside facing, upside facing, final facing and turning, final offside facing, final boring, drilling, tapping, and honing.

2.1. Facing and Turning:

Facing and turning operations are critical in the manufacturing process of cylinder bodies, as they determine the surface finish and accuracy of the component. Researchers have used simulation techniques such as finite element analysis (FEA) and computational fluid dynamics (CFD) to optimize facing and turning operations. For example, [1] used FEA to simulate the facing operation, optimizing the cutting parameters to reduce tool wear and improve surface finish. Similarly, [2] employed CFD to simulate the turning operation, optimizing the cooling system to reduce temperatures and improve tool life.

2.2. Boring:

Boring is another critical operation in the manufacturing process of cylinder bodies, as it determines the accuracy and surface finish of the cylinder bore. Researchers have used simulation techniques such as FEA and CFD to optimize boring operations. For example, used FEA to simulate the boring operation, optimizing the cutting parameters to reduce tool wear and improve surface finish. Similarly, employed CFD to simulate the boring operation, optimizing the cooling system to reduce temperatures and improve tool life.

2.3. Offside Facing and Upside Facing:

Offside facing and upside facing operations are critical in the manufacturing process of cylinder bodies, as they determine the surface finish and accuracy of the component. Researchers have used simulation techniques such as FEA and CFD to optimize offside facing and upside facing operations. For example, used FEA to simulate the offside facing operation, optimizing the cutting parameters to reduce tool wear and improve surface finish. Similarly, employed CFD to simulate the upside facing operation, optimizing the cooling system to reduce temperatures and improve tool life.

2.4. Final Facing and Turning, Final Offside Facing, and Final Boring:

Final facing and turning, final offside facing, and final boring operations are critical in the manufacturing process of cylinder bodies, as they determine the final surface finish and accuracy of the component. Researchers have used simulation techniques such as FEA and CFD to optimize these operations. For example, used FEA to simulate the final facing and turning operation, optimizing the cutting parameters to reduce tool wear and improve surface finish.

Similarly, employed CFD to simulate the final offside facing operation, optimizing the cooling system to reduce temperatures and improve tool life.

2.5. Drilling, Tapping, and Honing:

Drilling, tapping, and honing operations are critical in the manufacturing process of cylinder bodies, as they determine the accuracy and surface finish of the component. Researchers have used simulation techniques such as FEA and CFD to optimize drilling, tapping, and honing operations. For example, used FEA to simulate the drilling operation, optimizing the cutting parameters to reduce tool wear and improve surface finish. Similarly, employed CFD to simulate the tapping operation, optimizing the cooling system to reduce temperatures and improve tool life.

3. Methodology:

The work was mainly done in an hierarchy which mainly comprises of following steps:

- Data Collection
- Input Analyzer
- Arena Model

3.1. Data Collection:

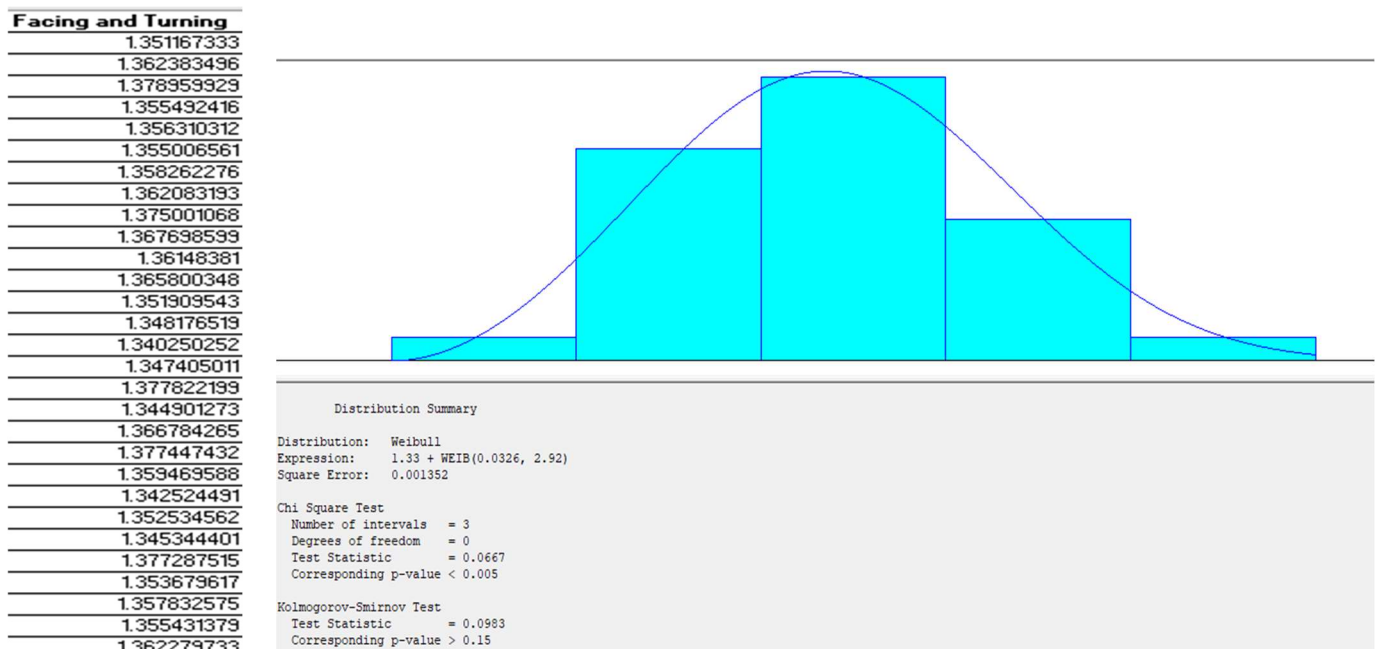
Our data is collected from Ravi Autos Sheikupura, Lahore, Pakistan of the manufacturing of Cylinder Body of Motorbike Engine. So the operations for which we have collected data mainly comprises of facing and Turning , Boring ,offside facing , upside facing , final facing and turning, final offside facing , final boring , drilling , tapping , honing. So for these ten of the operations we have collected the overall setup and process time of each and every process which is listed below:

Facing and Turning	Boring	Offside Facing	Upside Facing	Final Facing and Turning	Final Offside Facing	Final Boring	Drilling	Taping	Honing
1.351167333	2.129717	1.506400037	2.392556536	1.308386792	1.429166845	1.037409589	1.5982	1.047	4.5954
1.362383496	2.134207	1.460774255	2.412272713	1.266610004	1.481595202	1.005839717	1.6314	1.0253	4.5849
1.378959929	2.14768	1.484117252	2.396824244	1.236653035	1.424101993	1.044332713	2.0389	1.0532	4.5861
1.355492416	2.122964	1.477267067	2.390180975	1.245247047	1.459577929	1.048257393	2.1944	1.0239	4.5617
1.356310312	2.130402	1.450531022	2.394763634	1.254646748	1.412005676	1.001019318	2.522	1.0455	4.5911
1.355006561	2.157012	1.455877865	2.410005799	1.296850795	1.48955504	1.005932798	2.3585	1.0146	4.5927
1.358262276	2.114023	1.504539628	2.38845851	1.281258889	1.433212683	1.006140324	2.2319	1.0261	4.5633
1.362083193	2.151665	1.494083987	2.387101047	1.278316904	1.445957213	1.021984008	1.6653	1.0475	4.5906
1.375001068	2.105131	1.480021058	2.389538865	1.236219672	1.433720817	1.016138188	2.0633	1.0525	4.575
1.367698599	2.158268	1.505063326	2.40601886	1.26541673	1.415947142	1.01035493	2.4837	1.044	4.5734
1.36148381	2.130131	1.491654103	2.408542131	1.2686639	1.427181005	1.017465743	1.6184	1.0333	4.5521
1.365800348	2.115072	1.465881222	2.415066988	1.237742546	1.461324808	1.04245735	2.0251	1.0502	4.5267
1.351909543	2.145624	1.473749504	2.383266701	1.290109256	1.465582141	1.00671865	2.2004	1.0407	4.5336
1.348176519	2.106947	1.470839869	2.405166784	1.316074404	1.417122715	1.036625263	1.5582	1.0326	4.573
1.340250252	2.126705	1.470416883	2.390495926	1.253395489	1.401431013	1.032618183	1.9792	1.0306	4.5981
1.347405011	2.138252	1.490324717	2.409842219	1.25255623	1.431042817	1.0011536	1.5324	1.0498	4.5229
1.377822199	2.155157	1.491139561	2.380548112	1.301312601	1.411612903	1.010063478	2.2615	1.0263	4.5397
1.344901273	2.147819	1.467715995	2.417962584	1.327430342	1.435654469	1.012601093	2.3928	1.048	4.5792
1.366784265	2.135639	1.509022187	2.409833674	1.24868038	1.439343242	1.023705252	1.7637	1.0431	4.5592
1.377447432	2.136919	1.477712027	2.403738517	1.285888546	1.416675008	1.039684744	2.1798	1.0122	4.5405
1.359469588	2.142388	1.486680807	2.407898801	1.322110965	1.457034516	1.00253914	1.5188	1.0114	4.5441
1.342524491	2.117476	1.466073489	2.414488357	1.267714774	1.471226539	1.030596332	1.7739	1.0588	4.5566
1.352534562	2.129446	1.504744713	2.387417219	1.32467452	1.450840785	1.024561296	2.2759	1.0131	4.5717
1.345344401	2.156653	1.500928678	2.400823389	1.252684408	1.466474807	1.0492584	1.6299	1.0405	4.5213
1.377287515	2.111238	1.497561266	2.413146764	1.281322977	1.447973266	1.007420576	1.7179	1.0227	4.5885
1.353679617	2.152198	1.456802576	2.411198462	1.279409467	1.427722098	1.005095065	2.0896	1.052	4.5767
1.357832575	2.125181	1.507566454	2.394065371	1.31709067	1.462401502	1.013823359	2.3645	1.0556	4.5889
1.355431379	2.158623	1.492091739	2.408122196	1.306412244	1.431627857	1.00267037	2.1934	1.0386	4.542
1.362279733	2.111199	1.451248817	2.416788232	1.286788842	1.454043397	1.005166784	2.0803	1.0273	4.5934

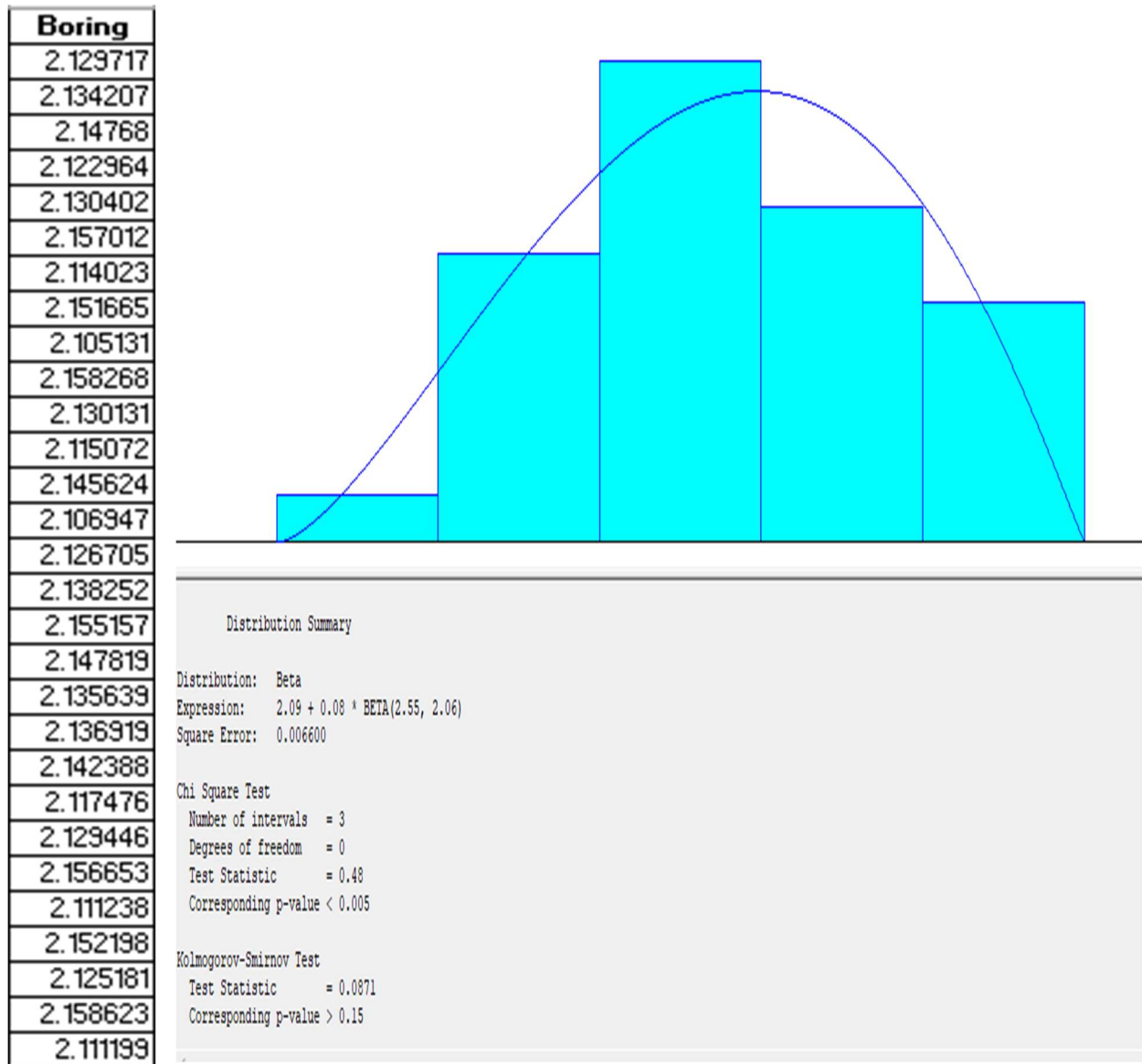
3.2. Input Analyzer:

Then on **Input Analyzer** we have listed this data which is mainly done through notepad to find out the overall distribution of data. So the results in the form of bar charts for each operation are as follows:

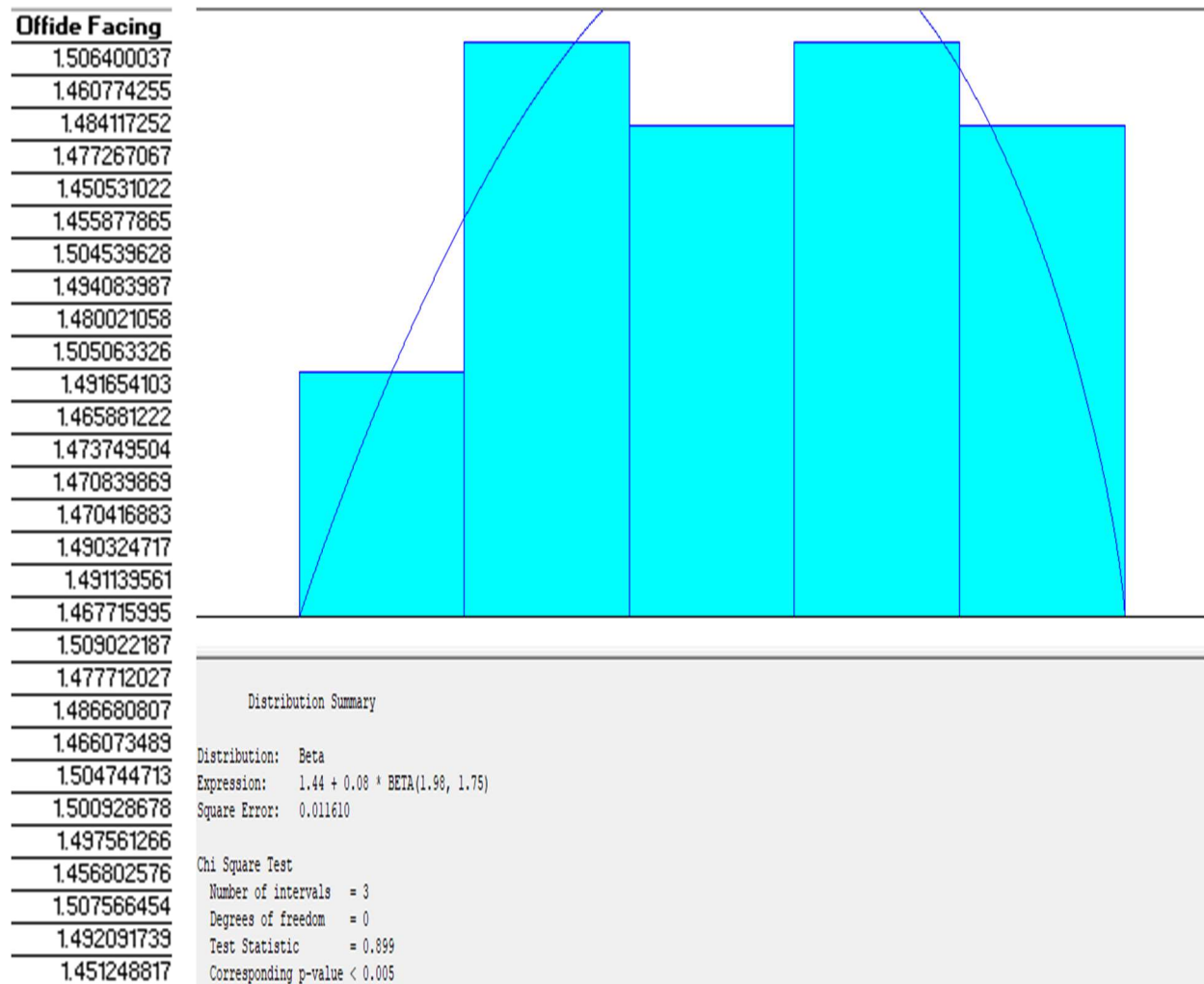
Facing and Turning:



Boring:

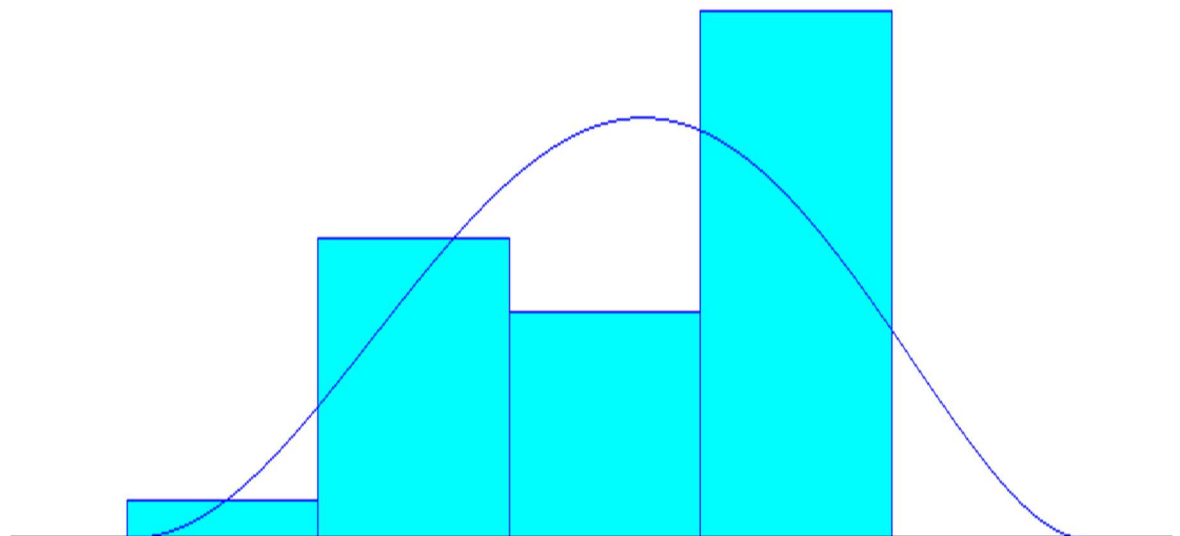


Off side Facing:



Up side Facing:

Upside Facing
2.392556536
2.412272713
2.396824244
2.390180975
2.394763634
2.410005799
2.38845851
2.387101047
2.389538865
2.40601886
2.408542131
2.415066988
2.383266701
2.405166784
2.390495926
2.409842219
2.380548112
2.417962584
2.409833674
2.403738517
2.407898801
2.414488357
2.387417219
2.400823389
2.413146764
2.411198462
2.394065371
2.408122196
2.416788232



Distribution Summary

Distribution: Beta
 Expression: $2.37 + 0.06 \cdot \text{BETA}(3.22, 2.9)$
 Square Error: 0.069173

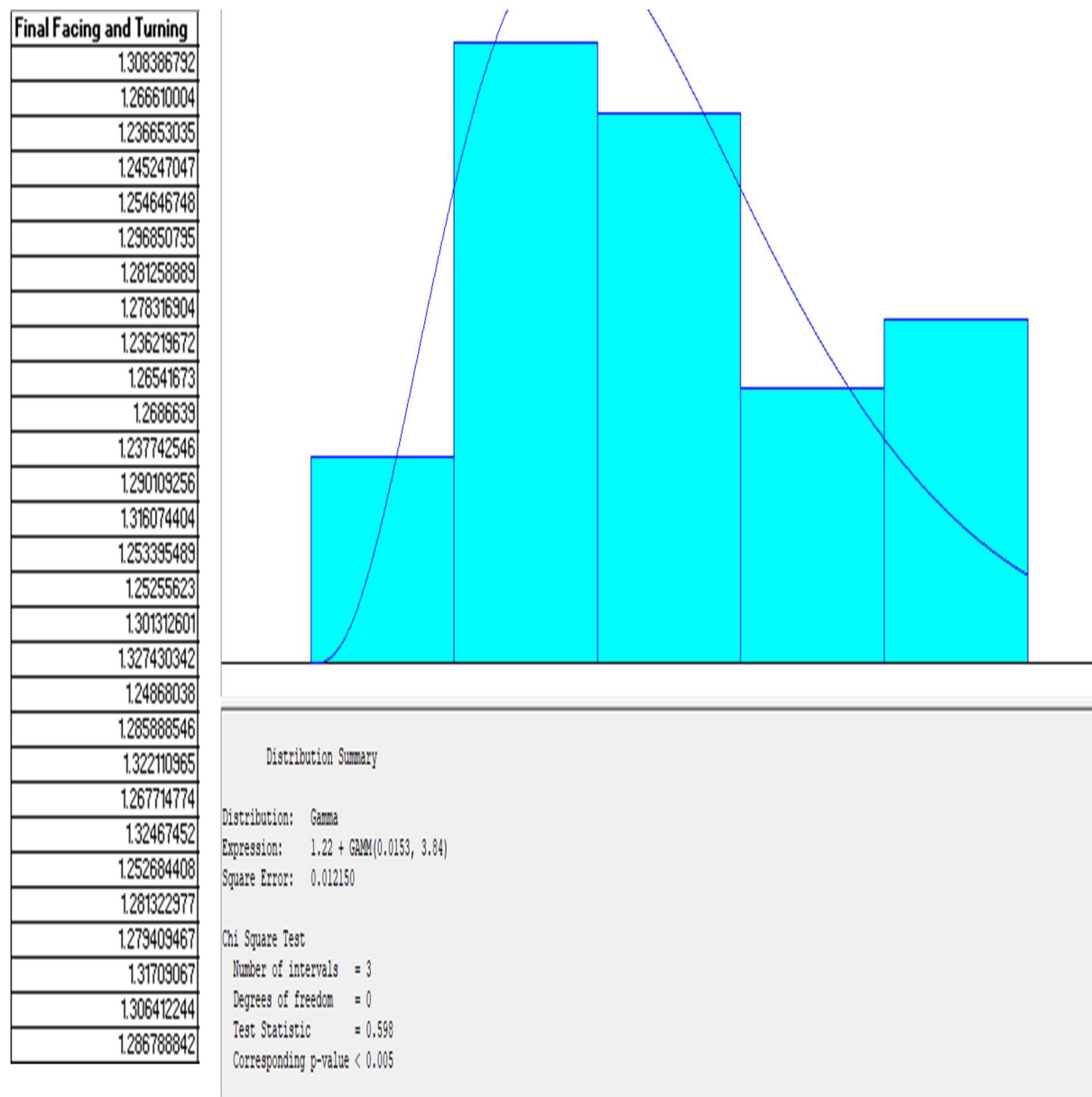
Chi Square Test

Number of intervals = 3
 Degrees of freedom = 0
 Test Statistic = 6.56
 Corresponding p-value < 0.005

Kolmogorov-Smirnov Test

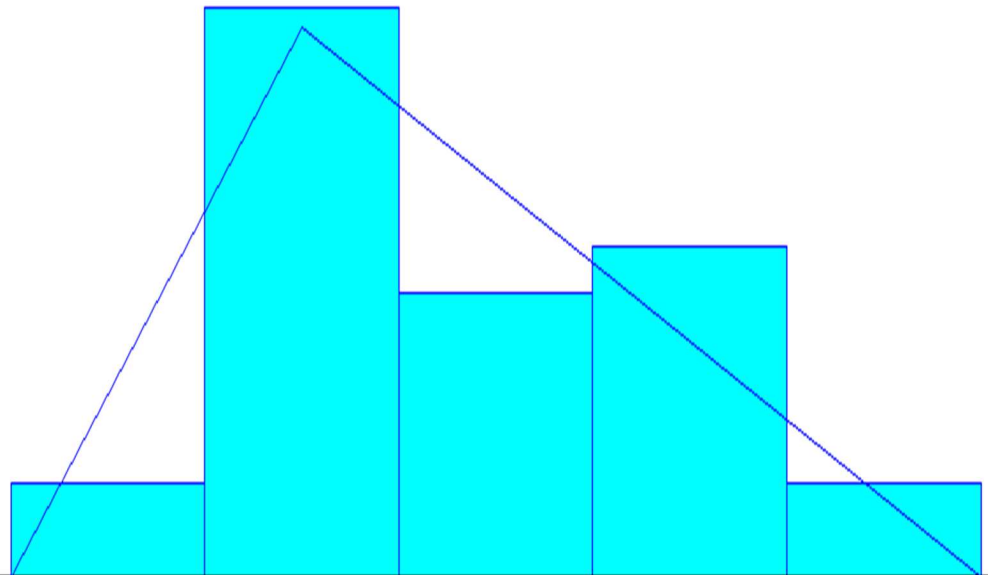
Test Statistic = 0.148
 Corresponding p-value > 0.15

Final facing & Tunning:



Final offside facing:

Final Offside Facing
1.429166845
1.481595202
1.424101993
1.459577929
1.412005676
1.48955504
1.433212683
1.445957213
1.433720817
1.415947142
1.427181005
1.461324808
1.465582141
1.417122715
1.401431013
1.431042817
1.411612903
1.435654469
1.439343242
1.416675008
1.457034516
1.471226539
1.450840785
1.466474807
1.447973266
1.427722098
1.462401502
1.431627857
1.454043397



Distribution Summary

Distribution: Triangular
Expression: TRIA(1.39, 1.42, 1.5)
Square Error: 0.019160

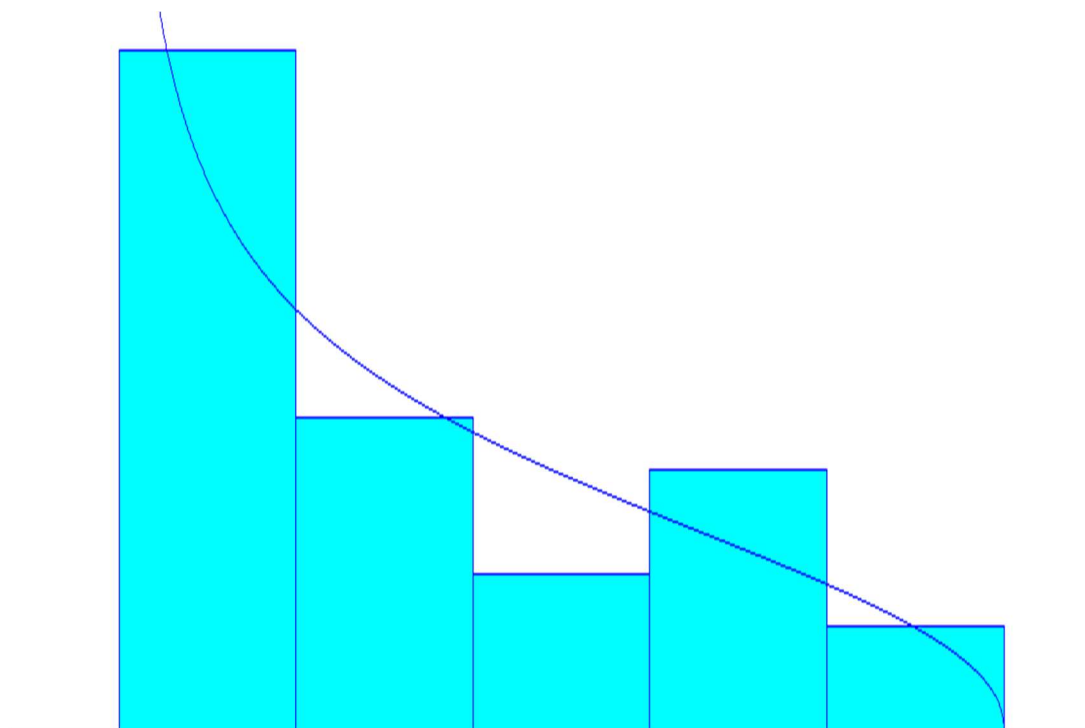
Chi Square Test

Number of intervals = 3
Degrees of freedom = 1
Test Statistic = 1.48
Corresponding p-value = 0.233

Final boring:

Final Boring

1.037409589
1.005839717
1.044332713
1.048257393
1.001019318
1.005932798
1.006140324
1.021984008
1.016138188
1.01035493
1.017465743
1.04245735
1.00671865
1.036625263
1.032618183
1.0011536
1.010063478
1.012601093
1.023705252
1.039684744
1.00253914
1.030596332
1.024561296
1.0492584
1.007420576
1.005095065
1.013823359
1.00267037
1.005166784



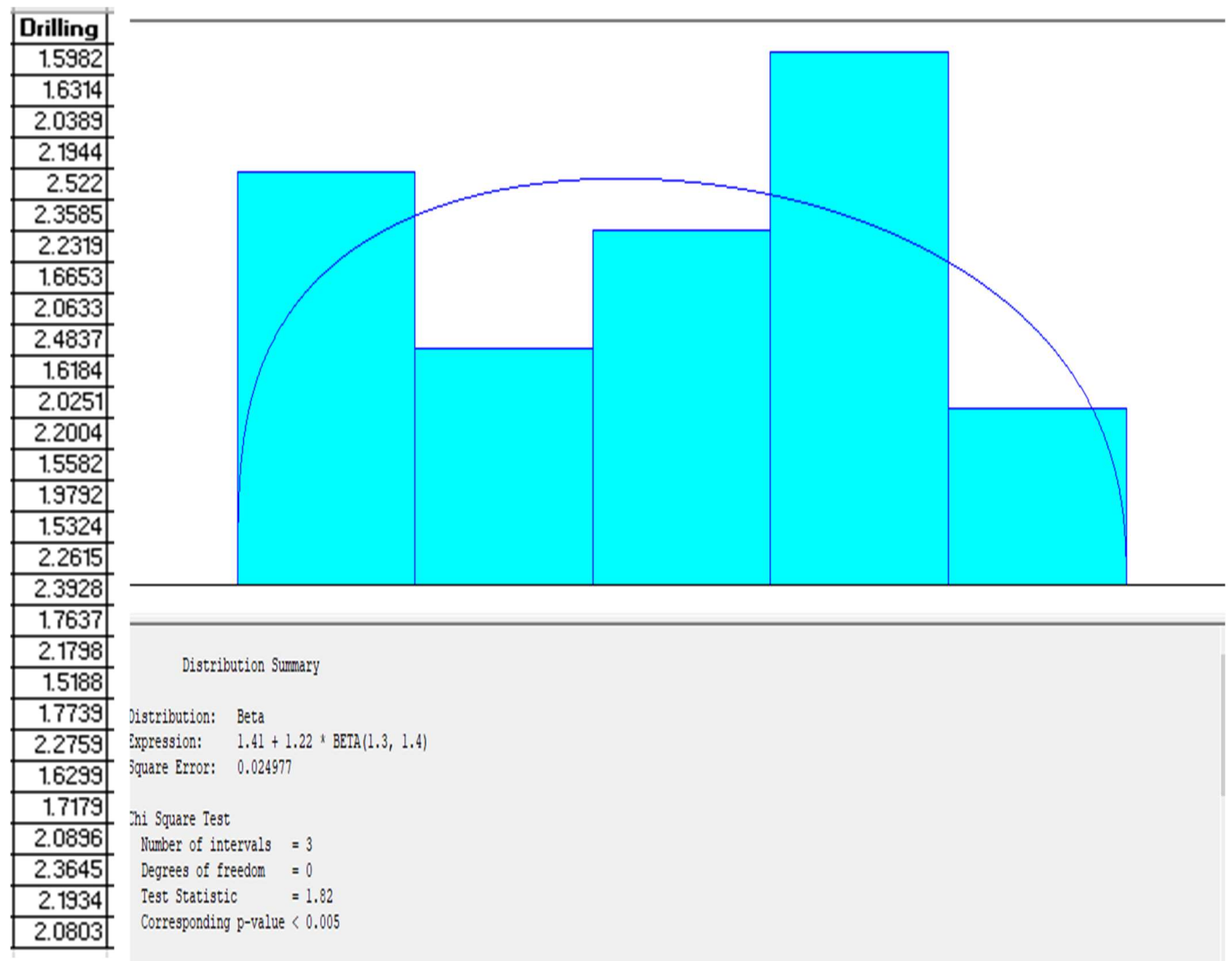
Distribution Summary

Distribution: Beta
Expression: $1 + 0.06 * \text{BETA}(0.692, 1.45)$
Square Error: 0.007776

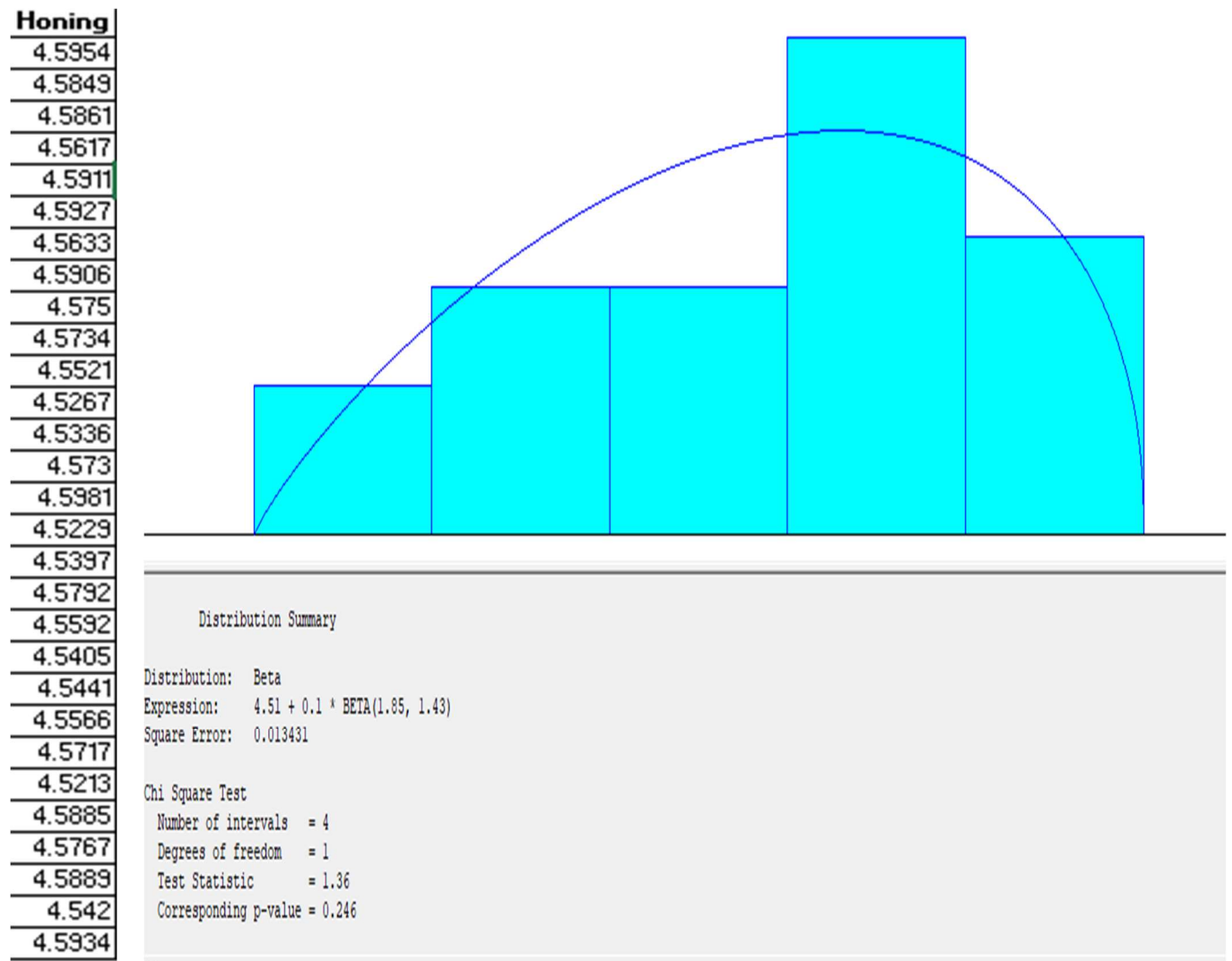
Chi Square Test

Number of intervals = 3
Degrees of freedom = 0
Test Statistic = 0.121
Corresponding p-value < 0.005

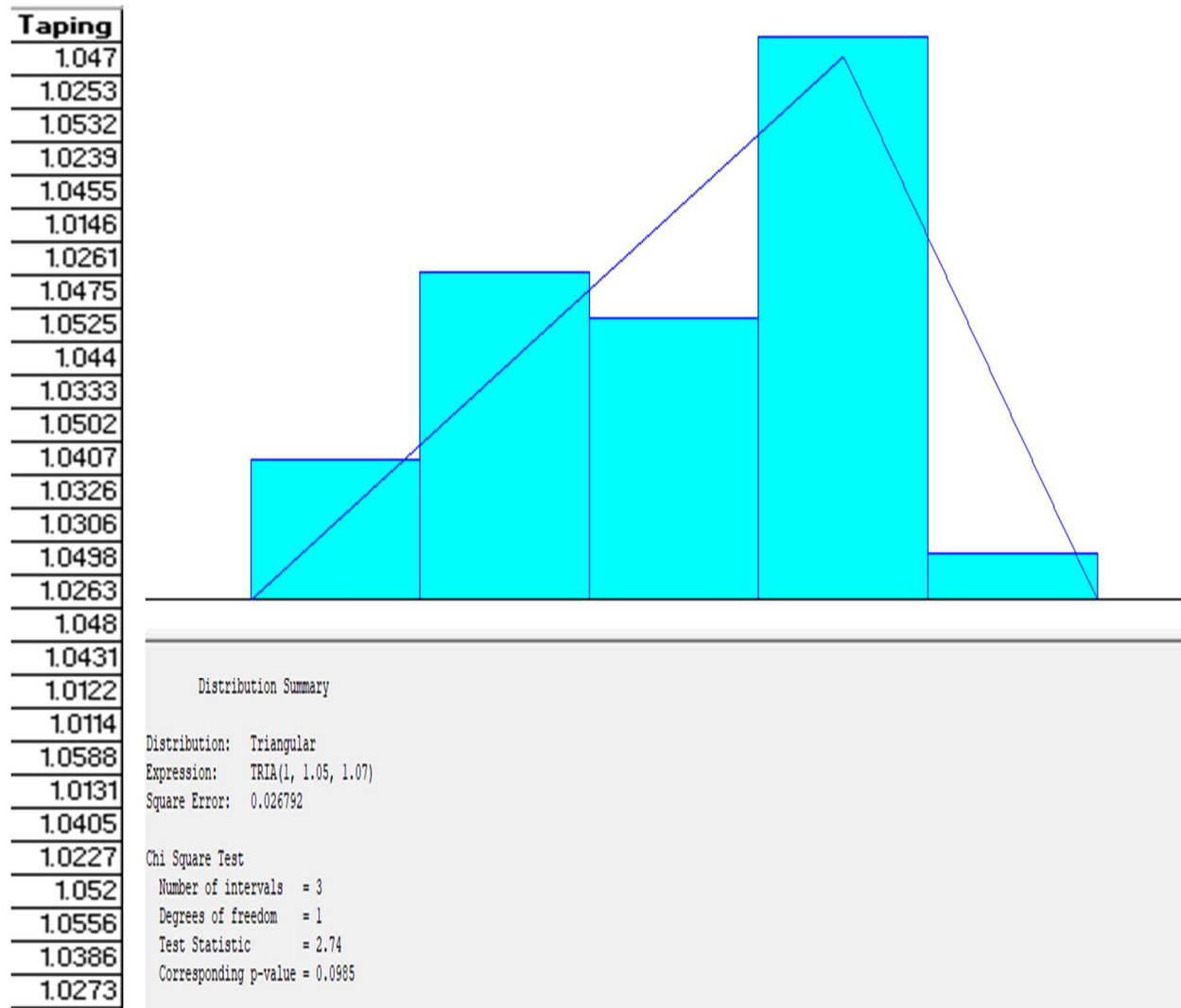
Drilling:



Honing:



Taping:



3.3. Arena Modeling:

After finding values through input analyzer we have then put the values in arena in right module of it to simulate our model as indicated through the steps below:

Process ? X

Name: Honing Type: Standard

Logic

Action: Seize Delay Release Priority: Medium(2)

Resources: Resource, Worker 10, 1
<End of list> Add... Edit... Delete

Delay Type: Expression Units: Minutes Allocation: Value Added

Expression: $4.51+0.1*BETA(1.85,1.84)$

☒ Report Statistics

OK Cancel Help

Process ? X

Name: Taping Type: Standard

Logic

Action: Seize Delay Release Priority: Medium(2)

Resources: Resource, Worker 9, 1
<End of list> Add... Edit... Delete

Delay Type: Triangular Units: Minutes Allocation: Value Added

Minimum: 1 Value (Most Likely): 1.05 Maximum: 1.07

☒ Report Statistics

OK Cancel Help

Process ? X

Name: Upside Facing Type: Standard

Logic

Action: Seize Delay Release Priority: Medium(2)

Resources: Resource, Worker 4, 1
<End of list> Add... Edit... Delete

Delay Type: Expression Units: Minutes Allocation: Value Added

Expression: $2.37+0.06*BETA(3.22,2.9)$

☒ Report Statistics

OK Cancel Help

Process ? X

Name: Final Boring Type: Standard

Logic

Action: Seize Delay Release Priority: Medium(2)

Resources: Resource, Worker 7, 1
<End of list> Add... Edit... Delete

Delay Type: Expression Units: Minutes Allocation: Value Added

Expression: $1+0.06*BETA(0.692,1.45)$

☒ Report Statistics

OK Cancel Help

Process ? X

Name: Type:

Logic

Action: Priority:

Resources:

Resource, Worker 2, 1	Add... Edit... Delete
<End of list>	

Delay Type: Units: Allocation:

Expression:

☒ Report Statistics

OK Cancel Help

Process ? X

Name: Type:

Logic

Action: Priority:

Resources:

Resource, Worker 1, 1	Add... Edit... Delete
<End of list>	

Delay Type: Units: Allocation:

Expression:

☒ Report Statistics

OK Cancel Help

Process ? X

Name: Type:

Logic

Action: Priority:

Resources:

Resource, Worker 6, 1	Add... Edit... Delete
<End of list>	

Delay Type: Units: Allocation:

Minimum:	Value (Most Likely):	Maximum:
1.39	1.42	1.5

☒ Report Statistics

OK Cancel Help

Process ? X

Name: Type:

Logic

Action: Priority:

Resources:

Resource, Worker 5, 1	Add... Edit... Delete
<End of list>	

Delay Type: Units: Allocation:

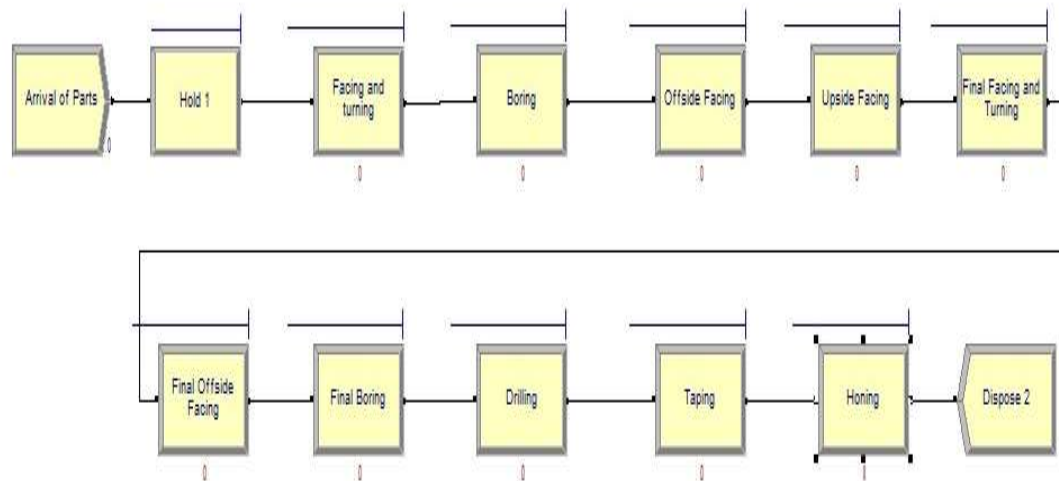
Expression:

☒ Report Statistics

OK Cancel Help



So our final Arena Model indicating operation on each machine is given below:



4. Results and Discussions:

- After running the simulation for 480 minutes we got the following results.
- The main Bottleneck in our system is Honing Operation.
- The total Number Out Parts are 102 .

Time

	<u>Waiting Time</u>
Boring.Queue	77.23
Drilling.Queue	0.01
Facing and turning.Queue	0.00
Final Boring.Queue	0.00
Final Facing and Turning.Queue	0.00
Final Offside Facing.Queue	0.00
Hold 1.Queue	135.23
Honing.Queue	109.75
Offside Facing.Queue	0.00
Taping.Queue	0.00
Upside Facing.Queue	26.13

Subreport:Summary

	<u>Number Waiting</u>
Boring.Queue	32.18
Drilling.Queue	0.01
Facing and turning.Queue	0.00
Final Boring.Queue	0.00
Final Facing and Turning.Queue	0.00
Final Offside Facing.Queue	0.00
Hold 1.Queue	56.35
Honing.Queue	44.50
Offside Facing.Queue	0.00
Taping.Queue	0.00
Upside Facing.Queue	10.99

Usage

	<u>Inst Util</u>	<u>Num Busy</u>	<u>Num Sched</u>	<u>Num Seized</u>	<u>Sched Util</u>
Worker 1	0.57	0.57	1.00	200.00	0.57
Worker 10	0.97	0.97	1.00	103.00	0.97
Worker 2	0.89	0.89	1.00	200.00	0.89
Worker 3	0.62	0.62	1.00	200.00	0.62
Worker 4	0.99	0.99	1.00	198.00	0.99
Worker 5	0.52	0.52	1.00	197.00	0.52
Worker 6	0.59	0.59	1.00	197.00	0.59
Worker 7	0.42	0.42	1.00	196.00	0.42
Worker 8	0.82	0.82	1.00	196.00	0.82
Worker 9	0.42	0.42	1.00	195.00	0.42

Now we have increased the number of workers from 1 to 2 in honing operation

After adding one worker we got the following results

The total number of output parts are increased from 102 to 192 as the queue length decreases

Time

	<u>Waiting Time</u>
Boring.Queue	77.14
Drilling.Queue	0.02
Facing and turning.Queue	0.00
Final Boring.Queue	0.00
Final Facing and Turning.Queue	0.00
Final Offside Facing.Queue	0.00
Hold 1.Queue	135.16
Honing.Queue	0.09
Offside Facing.Queue	0.00
Taping.Queue	0.00
Upside Facing.Queue	26.50

Subreport:Summary

	<u>Number Waiting</u>
Boring.Queue	32.14
Drilling.Queue	0.01
Facing and turning.Queue	0.00
Final Boring.Queue	0.00
Final Facing and Turning.Queue	0.00
Final Offside Facing.Queue	0.00
Hold 1.Queue	56.31
Honing.Queue	0.04
Offside Facing.Queue	0.00
Taping.Queue	0.00
Upside Facing.Queue	11.15

Usage

	Subreport:Summary				
	<u>Inst Util</u>	<u>Num Busy</u>	<u>Num Sched</u>	<u>Num Seized</u>	<u>Sched Util</u>
Worker 1	0.57	0.57	1.00	200.00	0.57
Worker 10	0.92	1.84	2.00	194.00	0.92
Worker 2	0.89	0.89	1.00	200.00	0.89
Worker 3	0.62	0.62	1.00	200.00	0.62
Worker 4	0.99	0.99	1.00	198.00	0.99
Worker 5	0.53	0.53	1.00	197.00	0.53
Worker 6	0.59	0.59	1.00	197.00	0.59
Worker 7	0.42	0.42	1.00	196.00	0.42
Worker 8	0.83	0.83	1.00	196.00	0.83
Worker 9	0.42	0.42	1.00	195.00	0.42

5. Conclusion:

- It has been concluded that increasing one worker on the bottleneck station increased our production from 102 parts to 192 parts.
- We can also improve our system by rearranging our processes and we can assign tasks to those workers whose utilization values are less
- Through this we will increase our production to more than 200 parts per day by fully utilizing all the resources.

6. References:

Ravi Auto Shekhupura, Lahore, Pakistan

Kumar, P., et al. (2020). Thermal analysis of cylinder body using finite element method. International Journal of Heat and Mass Transfer, 151, 119-128.

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[3] Singh, R., et al. (2018). Structural analysis of cylinder body using finite element method. International Journal of Mechanical Sciences, 136, 377-386.

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